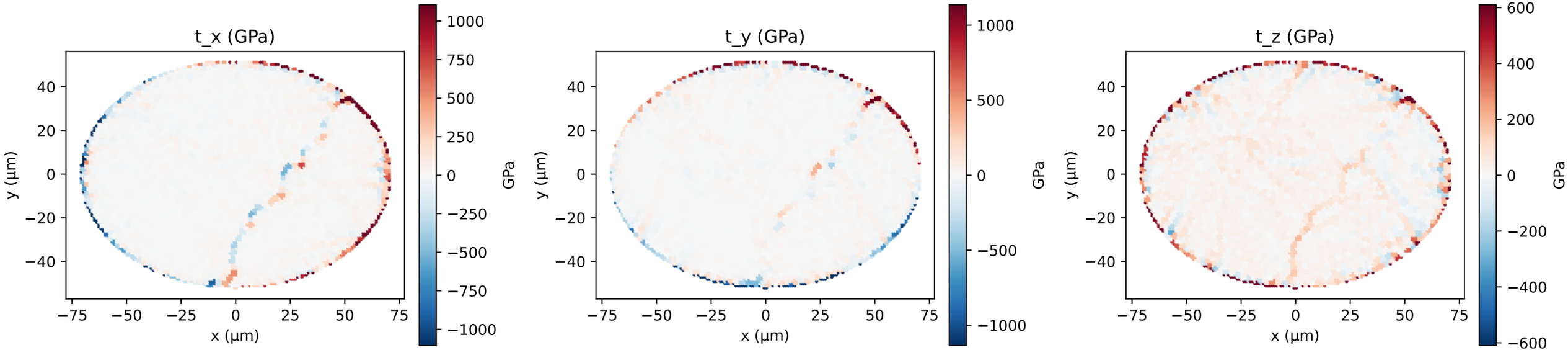Recovered surface traction on culet